

Reaction Mechanism Studies towards Effective Fabrication of Lithium-Rich Anti-Perovskite Li_3OX ($X = \text{Cl}, \text{Br}$)

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Abstract:

Lithium-rich Anti-perovskite (LiRAP), with general formula Li_3OX ($X = \text{Cl}, \text{Br}, \text{I}$), and recently reported as superionic conductors with 3-dimensional Li^+ migrating channels, is emerging as a promising candidate for solid electrolyte of all-solid-state LIBs. However, great challenge still remains to fabricate pure LiRAP due to the difficulty of the phase formation and moisture-sensitive nature of the products. In this work, we thoroughly studied the formation mechanism of Li_3OCl and Li_3OBr in various solid state reaction routes. Different experimental strategies were developed to improve the syntheses, in purposes of improved phase stability and large-scale production of LiRAP. One feasible method is to use strongly reductive agents Li metal or LiH to eliminate OH species. The results show that LiH is more effective than Li metal because of negatively charged H^- and uniform reaction. The other well-established method is using Li_2O and LiX mixture as reagents to preventing OH phase at the beginning, and using protected ball milling to make fine powders and hence active the reaction. IR spectroscopy, thermal analyses and first-principle calculation were performed to give indications on the reaction pathway.

Keywords: solid electrolyte, lithium ion battery, Lithium-rich Anti-Perovskite, solid state reaction

1. Introduction

Lithium ion batteries (LIBs) are important energy storage devices due to their wide application in electric vehicles, portable and stationary devices. [1-4] Conventional LIBs use lithium salts in organic solvents as electrolyte [5]. However, the liquid electrolytes are usually toxic, corrosive and flammable. [5,6] In addition, if using lithium metal as the anode, the dendrite resulted from Li^+ non-uniformed deposition either becomes “dead” lithium, reducing the battery efficiency, [7] or even worse breaks the separator, resulting in the battery short circuit. Recently developed solid state lithium ion batteries are expected to solve the safety and

environmental issues related to conventional LIBs. [8-11] The solid state LIBs use inorganic compounds as electrolytes, which are in-toxic, non-flammable, environmental friendly and safe. Furthermore, because of the intrinsic mechanical property, solid electrolytes can avoid the risk of lithium dendrite greatly. This enables solid state battery to use lithium metal as anode, and hence to maximum the working potential, capacity, power density and energy density. As the key component in solid state LiIBs, solid electrolyte needs to achieve some requirements such as high ionic conductivity, low conducting activation energy, negligible electronic conductivity, large electrochemical, and compatible with electrodes. [9] For instance, some inorganic compounds with Li^+ superionic conductivity are studied as solid electrolytes for LIB, showing good performances. [9,12-16] The newly developed lithium superionic conductor $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, with a new three-dimensional framework, has a Li^+ conductivity of 1.2×10^{-2} S/cm at room temperature, which is the highest ionic conductivity reported of solid electrolyte. [12] The garnet-type $\text{Li}_7\text{La}_3\text{ZrO}_{12}$ fast Li^+ conductor was observed to have a Li^+ 3-dimensional diffusion pathway. [13,14] Nanoporous $\beta\text{-Li}_3\text{PS}_4$ was reported to have conductivity of 1.6×10^{-4} S/cm at room temperature, wide electrochemical window and stability against Li metal. [15] NASICON [16] and LISICON [17,18] compounds are widely studied solid state fast lithium conductors these decades. Solid state thin film battery with LiPON electrolyte was demonstrated excellent performance with a high columbic efficiency of >99.98% and stable cycling for 10 000 cycles. [19]

Recently, a family of lithium-rich anti-perovskite (LiRAP), in the formula of Li_3OX ($X = \text{Cl}, \text{Br}, \text{I}$), was reported as Li^+ superionic conductor, a potential candidate for solid electrolyte in LiIBs. [20] The anti-perovskite comes from perovskite ABO_3 . NaMgF_3 was reported to have F^- superionic conductivity. [21] With electronic reversion, Li_3OX was expected to be a Li^+ superionic conductor. Li_3OX solid has *ca.* 60 at. % Li in lattices and light weighted. It has rich Li^+ and 3-D migrating channels. [22,23] Via cation- and/or anion-doping, Li^+ depleting, or other structural manipulation methods, the ionic conductivity of this class of materials could reach to 10^{-3} S/cm, and with low energy barrier. [20] Never the less, LiRAP has large electrochemical window, low electronic conductivity (the band gap of Li_3OCl exceeds 5 eV [22,23]). Li_3OCl was reported to be stable with Li metal, showing self-stabilization feature [24]. These properties make LiRAP a promising candidate as solid electrolyte for LiIBs.

As the starting point of battery research and development, it is essential to get pure and stable phase of LiRAP compounds via simple and economic synthesis pathway. In previous report, Li_3OX could be obtained by heating Li_2O and LiX mixture, but the yield was quite low. Another routine of Li_3OCl synthesis was realized by heating LiCl and LiOH mixture for a few days under vacuum in the purpose of removing H_2O . [20] Recently, Li_3OCl solid electrolyte films were fabricated via pulsed laser deposition (PLD) using Li_3OCl compound target. [24] LiRAP related materials were also synthesized via high-pressure high-temperature method [25], organic chemistry method [ref.], etc. However, none of these methods could meet the criterions for large scale industrial manufacture.

In this work, lithium rich Li_3OX ($X=\text{Cl}, \text{Br}$) was synthesized via solid state reaction with different approaches to get pure phase LiRAP on one hand to get rid of contamination of $-\text{OH}$ in the compounds and on the other hand to get high yield products economically. Two main routines were demonstrated: 1. reductive agents were adopted as starting materials, such as Li-metal and LiH, to remove $-\text{OH}$ that may exist in the products; 2. ball milling Li_2O and LiX raw materials to lower the kinetic barriers of reaction and enhance the productivity. Thermogravimetry, differential scanning calorimetry (TG-DSC) infrared spectroscopy (IR) and theoretical calculation were adopted to clarify the reaction mechanism.

2. Experimental

2.1 Sample Preparation

LiRAP samples were synthesized with powder Li_2O (Alfa Aesar, 99.5 %), LiCl (Alfa Aesar, anhydrous, 99 %), LiBr (Alfa Aesar, anhydrous, 99 %), LiOH (Alfa Aesar, anhydrous, 98 %), Li metal (Sigma) and LiH (Alfa Aesar, 97 %) as starting materials. The materials were weighted, mixed, grinded and heated inside glovebox ($\text{H}_2\text{O} < 5 \text{ ppm}$, $\text{O}_2 < 20 \text{ ppm}$). The mixture in crucibles was placed in a compact furnace inside glovebox for sintering. For ball milling process, the raw materials were loaded into sealed ball milling jars inside glovebox and then transferred to ball milling machine.

2.2 Sample characterization

The X-ray powder diffraction (XRD) patterns were recorded on an X-ray diffractometer (Bruker D8 Advance). The as synthesized powders were put on normal sample holder with parafilm on the surface for protection, or in a zero background air tight sample holder.

IR spectroscopy was collected on a Nicolet IS5 Mid-infrared FT-IR spectrometer.

Calorimetric property of the LiRAP was tested on Netzsch STA 449 C. Samples were sealed in alumina crucible with cap inside glovebox. Ar gas was used as carrier gas during test. TG-DSC measurements were recorded with heating/cooling rate as 10 K/min.

For conductivity measurement, the sample powders was melted and pressed between two pieces of gold foil before loaded into a sealed sample holder. The whole assemble was transferred to a tube furnace with protecting Ar gas. The electrochemical impedance spectroscopic (EIS) measurement was carried out on an Solartron 1260 instrument (10 mV perturbation voltage and between 1 MHz and 100 mHz) as function of temperature from 25 °C to 175 °C.

2.3 Theoretical calculation

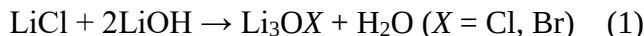
The crystal structures of Li_2O , LiCl and Li_3OCl from ICSD database were employed as the initial structure model to study the total energy as a function of cell volume. The calculations

were performed by using CASTEP software [26] at the GGA level of theory and energy cutoff of 380 eV. The total energies of the reactants ($\text{Li}_2\text{O} + \text{LiCl}$) were added by their individual formation energy with decreased or increased cell volumes with the same percentages.

3. Results and Discussion

3.1. Reductive agents as starting materials

As reported in our previous work [20], Li_3OCl was synthesized by heating LiCl and LiOH mixture under vacuum at 330 - 360 °C for several days, as shown in equation (1).



Long time heating under vacuum is necessary to remove water and drives the reaction going forward. However, this method has high requirements on vacuum condition and takes long time. Therefore, reductive agents were adopted to break the O-H bond and hence to promote the reaction going forward.

3.1.1. Lithium metal as a reducer

Lithium metal was adopted as reductive agent. It reacts with LiOH and LiX ($X = \text{Cl}, \text{Br}$) via equation (2). The starting materials, Li metal, LiOH and LiX , were mixed by 1.05:1:1 molar ratio, where 5 mol. % excess Li metal was added as the evaporation lost during calcination. The mixture was heated at 600 °C for 10 hours under vacuum. After cooling down, the calcinated product was grinded and heated multi-times to get pure phase anti-perovskite Li_3OX . Fig. 1 (a) shows the XRD pattern of the as synthesized Li_3OCl . It was indexed as the cubic $Pm-3m$ phase, with lattice parameter $a = 3.9161 \text{ \AA}$.

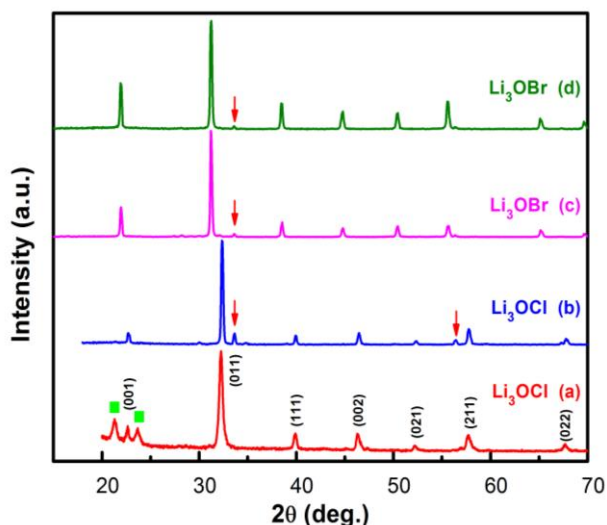
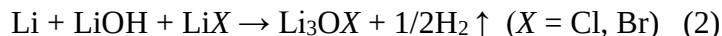


Fig. 1 Powder X-ray diffraction pattern of Li_3OCl synthesized via $\text{Li} + \text{LiOH} + \text{LiCl} \rightarrow \text{Li}_3\text{OCl} + 1/2\text{H}_2 \uparrow$ (a), Li_3OCl via $\text{LiH} + \text{LiOH} + \text{LiCl} \rightarrow \text{Li}_3\text{OCl} + \text{H}_2 \uparrow$ (b), Li_3OBr via $\text{LiH} + \text{LiOH} + \text{LiBr} \rightarrow \text{Li}_3\text{OBr} + \text{H}_2 \uparrow$ (c), and Li_3OBr via ballmilling and sintering methode $\text{Li}_2\text{O} + \text{LiBr} \rightarrow \text{Li}_3\text{OBr}$ (d). Indexing was performed by using space group $Pm-3m$, $a = 3.9161 \text{ \AA}$ for Li_3OCl and $a = 4.0468 \text{ \AA}$ for Li_3OBr . Peaks marked by solid green square are contributed by parafilm, and by red arrow are ascribed to Li_2O .

For this method, there are different possible reaction pathways. First possibility is that lithium metal reacts with LiOH to form Li_2O and release H_2 ; then highly activated Li_2O reacts with LiX to form Li_3OX , as shown in equation (3) and (4). Second is that LiOH and LiX react together, and then the product $\text{Li}_2(\text{OH})\text{X}$ reacts with the rest Li metal, as shown in equation (5) and (6).

To clarify the reaction mechanism, thermal analysis was adopted to find the reaction routine. **Fig. 2** shows the TG-DSC curves of $\text{Li} + \text{LiOH}$ (a), $\text{Li} + \text{Li}_2(\text{OH})\text{Cl}$ (b), $\text{Li} + \text{LiOH} + \text{LiCl}$ (c), $\text{Li}_2\text{O} + \text{LiCl}$ (d) mixtures, and DSC curves of Li_3OCl and $\text{Li}_2(\text{OH})\text{Cl}$ (e). It is clearly that for the reaction of LiOH and Li metal (written as equation (3)), after Li melting at 179°C , the Li and LiOH react greatly at around 460°C , indicated by the exothermal peak and weight loss (H_2 release) in **Fig. 2** (a). However, the reaction of $\text{Li} + \text{LiOH} + \text{LiCl}$ mixture behaves differently (**Fig. 2** (b)). It started absorb energy from 273°C , and then release energy and H_2 gas greatly around 300°C . In DSC curve, the position of this exothermic peak is at 303°C . Compared with reaction of $\text{Li} + \text{LiOH}$ mixture, the existence of LiCl changes the reaction pathway and makes the reaction happens earlier. This result excludes the first assumption mentioned above.

Further thermal analysis on pure $\text{Li}_2(\text{OH})\text{Cl}$ and its mixture with Li -metal gives more indications about the reaction pathway. In TG-DSC curves (**Fig. 2c**), the thermal behavior of $\text{Li} + \text{Li}_2(\text{OH})\text{Cl}$ mixture was similar to that of the $\text{Li} + \text{LiOH} + \text{LiCl}$ mixture. It started absorbing energy at 270°C , This endothermic peak is around 285°C , corresponding to the melting temperature of $\text{Li}_2(\text{OH})\text{Cl}$, as shown clearly in DSC curves of Li_3OCl and $\text{Li}_2(\text{OH})\text{Cl}$ in **Figure 2 (e)**. Then the $\text{Li} + \text{Li}_2(\text{OH})\text{Cl}$ mixture release energy and H_2 gas greatly around 300°C , with the exothermic peak at 296°C .

The thermal analyses reveal the reaction mechanism of the three-reagent-reaction. With Li metal, LiOH and LiCl as starting materials, as temperature increase, Li metal melts around 180°C . Then LiOH and LiCl form $\text{Li}_2(\text{OH})\text{Cl}$ and melts. At the meantime, Li and $\text{Li}_2(\text{OH})\text{Cl}$ react quickly, to form Li_3OCl and H_2 , and release energy, confirming the second reaction pathway, $\text{Li}_2(\text{OH})\text{Cl}$ forming from LiCl and LiOH firstly and then reacting with Li metal, is the right assumption.

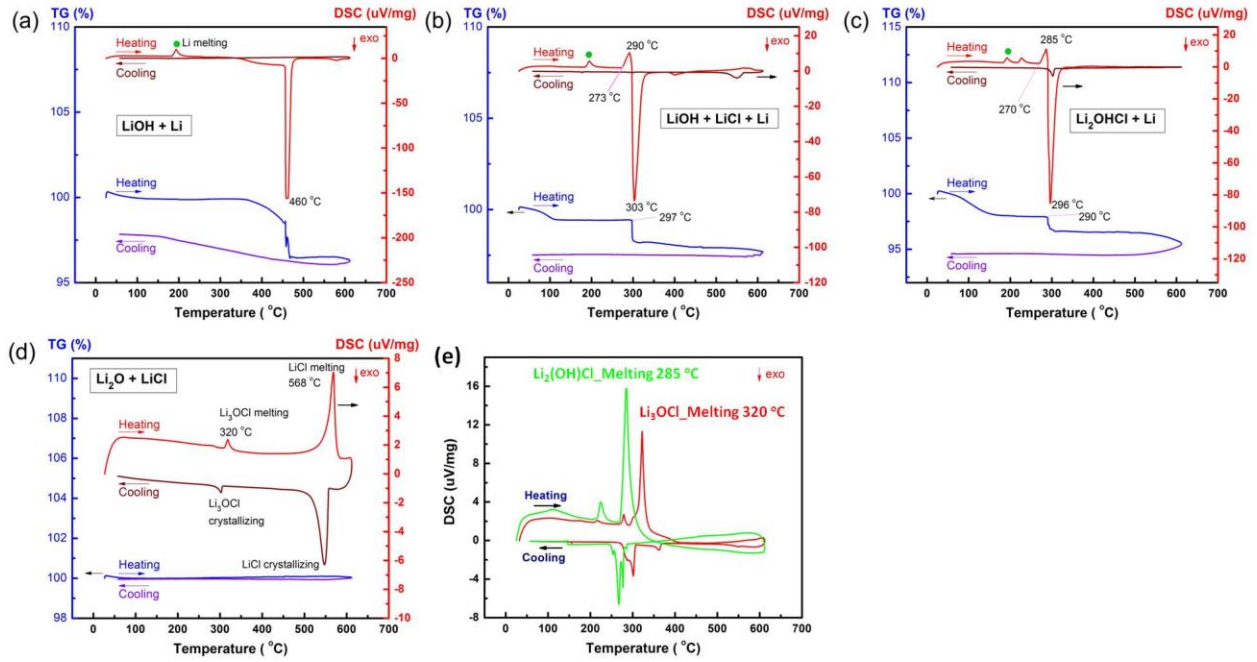
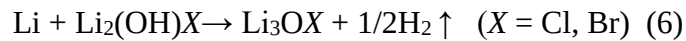
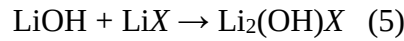
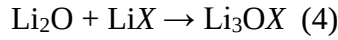
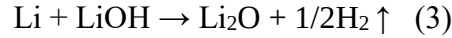


Fig. 2 TG-DSC curves of Li + LiOH (a), Li + LiOH + LiCl (b), Li + Li₂(OH)Cl (c) and Li₂O + LiCl (d) mixtures, DSC curves of Li₃OCl and Li₂(OH)Cl samples (e). The solid green circles at 180 °C in DSC curve in a, b and c represent Li-metal melting.



Using Li metal as one starting material is an effective method for Li₃OX (X=Cl, Br) synthesis. However, the reagents were not mixed uniformly, as the soft and light nature of lithium metal and lithium liquid resulting in uniform reaction. More important, not all the –OH group can be removed efficiently. As demonstrated by the IR spectra (**Fig. 3**), which is sensitive to the vibration of –OH group, some Li₃OCl batch, indexed to be cubic anti-perovskite by XRD, has strong O-H vibration modes. [27,28] The existence of OH impure phase results from uncontrolled aspects in synthesis process.

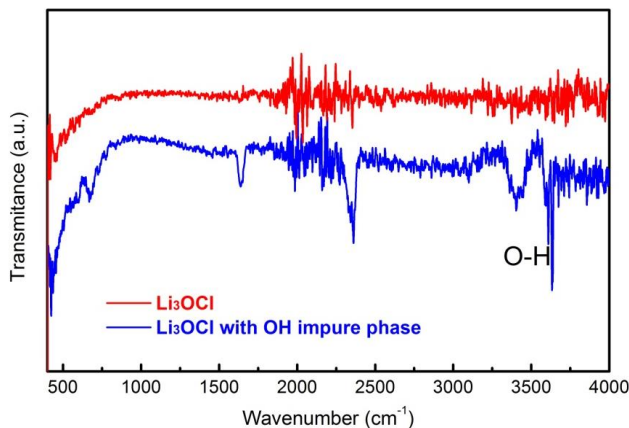
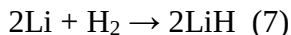


Fig. 3 IR spectra of Li_3OCl synthesized from Li metal with or without OH phase.

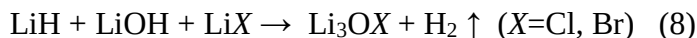
This synthesis could be further improved by using Li getter, which is additional Li metal placed beside the crucible containing starting materials. Although the additional piece of Li metal does participate in the reaction in equation (2), it could react with the generated H_2 to form LiH (equation 7 as below), driving the reaction 2 going forward.



As the non-uniform of the lithium metal and not repeatable reaction routine, the next reaction is to replace Li metal by LiH fine powder, as discussed below.

3.1.2. LiH as a reducer

With LiH as reducer to get Li_3OX ($X=\text{Cl}, \text{Br}$), the reaction could be expressed as:



The advantage of using LiH is that firstly the H^- in LiH is negatively charged, therefore it could break the O-H bond effectively; secondly, LiH is fine powder, so all the starting materials could be mixed uniformly and hence react thoroughly. For this reaction, the LiH , LiOH and LiX ($X=\text{Cl}, \text{Br}$) were mixed with 1:1:1 molar ratio and heated at 480°C for 1 hour. The product contains Li_3OX and tiny amount of Li_2O , LiX leftover. It should be noted that repeated grinding and heating is needed to get pure phase Li_3OX . The lattice parameters of synthesized Li_3OCl and Li_3OBr (Fig. 1 (b) (c)) are $3.92 \text{ \AA}$ and $4.05 \text{ \AA}$, respectively.

Similarly, this reaction could be separated into two steps referring to using lithium metal. The first step is the reaction of the LiOH and LiX to get $\text{Li}_2(\text{OH})\text{X}$, and the second step is the reaction of the obtained $\text{Li}_2(\text{OH})\text{X}$ with LiH , as shown in equation (9) and (10). There are several

merits of this routine. Firstly, in each step the two agents could be mixed uniformly and contact tightly. Secondly, it is easy to get $\text{Li}_2(\text{OH})\text{X}$, by heating LiOH and LiX mixture at 400 °C for 3 hours. Thirdly, the first step product is anti-perovskite ($\text{Li}_2(\text{OH})\text{Br}$, cubic) or distorted anti-perovskite ($\text{Li}_2(\text{OH})\text{Cl}$, orthorhombic). It is easy for them to transform to Li_3OX of the same structure. The XRD patterns of Li_3OX via this two-step reaction are shown in Fig. 4. The insets of magnified reflections around $2\theta = 32^\circ$ show the difference of $\text{Li}_2(\text{OH})\text{X}$ and Li_3OX . Orthorhombic $\text{Li}_2(\text{OH})\text{Cl}$ transforms to cubic Li_3OCl ; and cubic $\text{Li}_2(\text{OH})\text{Br}$ changes to cubic Li_3OBr with lattice expansion.

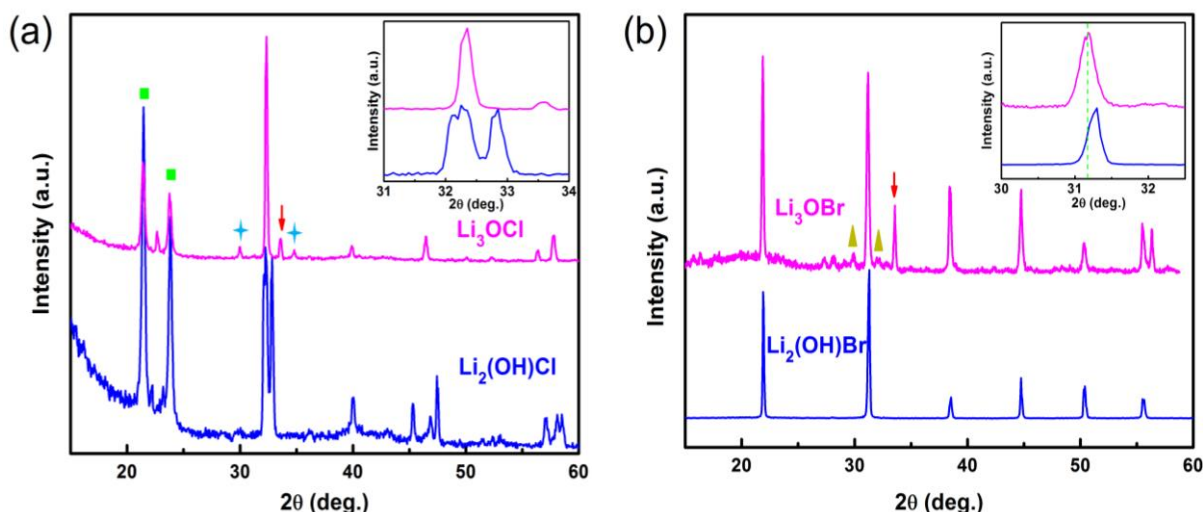
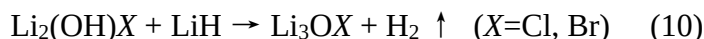


Fig. 4 XRD patterns of $\text{Li}_2(\text{OH})\text{X}$ and Li_3OX synthesized via $\text{LiOH} + \text{LiX} \rightarrow \text{Li}_2(\text{OH})\text{X}$ and $\text{Li}_2(\text{OH})\text{X} + \text{LiH} \rightarrow \text{Li}_3\text{OX} + \text{H}_2 \uparrow$, $\text{X}=\text{Cl}$ (left), Br (right). Insets are magnified reflections around $2\theta = 32^\circ$. Peaks noted with green square are contributed by parafilm, red arrow by Li_2O , blue star by LiCl and yellow triangle by LiBr .

3.2. Improved synthesis of Li_3OX from lithium oxide and lithium halides

Li_3OX could be formed from Li_2O and LiX ($\text{X}=\text{Cl}, \text{Br}$) directly, via reaction (11), as started at the beginning, the low yield is fatal even though avoiding H_2O and other OH phase from the very beginning.



Fig. 2 (d) gives the TG-DSC curves of Li_2O and LiCl mixture. The main peaks around 570 °C represent the melting and crystallizing of LiCl , and the tiny peak around 320 °C corresponds to

the melting and crystallizing of Li_3OCl . The low yield can be well explained by the theoretical calculation, as shown in Fig. 5, the energy difference between Li_3OCl and the starting material (Li_2O and LiCl) is quite small, and the formation energy of Li_3OCl is 5.01 kJ/mol higher than that of the starting materials ($\text{Li}_2\text{O} + \text{LiCl}$), indicating that Li_3OCl is not thermal stable and easily decomposes to Li_2O and LiCl . However, Li_3OCl can be kinetically obtained as indicated by and the thermal analysis.

The next step is to enhance the yield through kinetic control. Looking into the details of the reaction, Li_3OX is formed at the contact interface between Li_2O and LiX particles. Since the melting temperature of Li_3OX (320 °C) is much lower than those of Li_2O (1438 °C), LiCl (605-614 °C) and LiBr (552 °C), therefore, at the synthesizing temperature (*ca.* 500 °C), the as formed Li_3OX is melting and coating on the surface of starting material particles. Eventually, Li_3OX liquid blocks the further reaction between Li_2O and LiX , resulting in low yield. At this condition, grinding helps to expose fresh Li_2O surface, and promotes reaction. Repeat of grinding and heating is necessary to make the reaction completed.

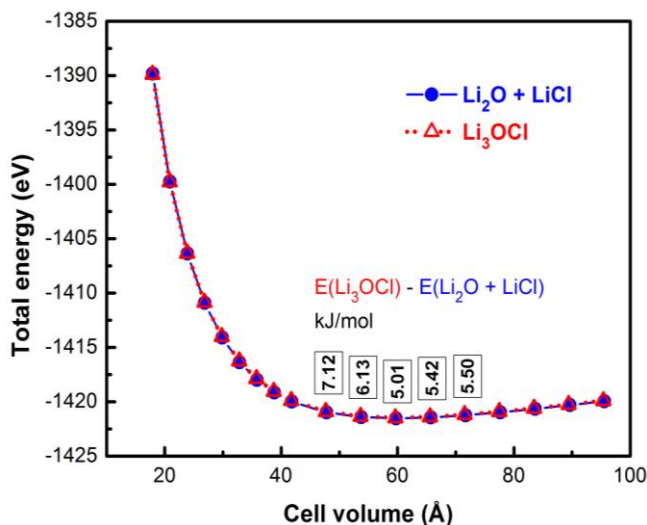


Fig. 5 Energy difference between lithium-rich anti-perovskite Li_3OCl and the starting materials ($\text{Li}_2\text{O} + \text{LiCl}$) as function of cell volume. Insets are energy differences around the energy minimum.

Based on the experiments and analyses above, a protected ball milling process was adopted to improve the synthesis. The ball milling of Li_2O and LiX mixture results in the fine powder or even nanosized powder raw material and uniformly mixing. The ball milled raw materials are more active and contacting thoroughly. Consequently, the reaction is promoted forward greatly, pure phase Li_3OBr could be obtained with only one time calcinating. XRD pattern in Fig. 1 (d) shows the pure Li_3OBr product via ball milling. It was indexed as the cubic $Pm-3m$ phase, with

lattice parameter $a = 4.0468 \text{ \AA}$. It should be mentioned that excess LiBr was added in raw materials considering the evaporating during sintering.

The conductivity of the as prepared LiRAP samples was investigated by EIS, and equivalent circuit was used to resolve the spectra (Fig. 6). In the studied temperature range ($25 \text{ }^{\circ}\text{C} - 175 \text{ }^{\circ}\text{C}$), the conductivity of Li_3OBr (obtained from LiBr and Li_2O) obeys the Arrhenius equation, with the action energy E_a as 0.74 eV , and conductivity as $1.0 \times 10^{-6} \text{ S/cm}$ at $25 \text{ }^{\circ}\text{C}$ (Fig. 6 (a)). In this work, samples synthesized via different methods give similar conductivity results. This conductivity value is lower than reported conductivity of Li_3OCl , which is $0.85 \times 10^{-3} \text{ S/cm}$ at room temperature. [20] It should be mentioned that lower conductivities of LiRAP family were also reported by other researchers. [24] In this work, it probably arises from sample synthesize and fabrication processes. In Fig. 6 (b), the typical impedance spectra of Li_3OBr at $25 \text{ }^{\circ}\text{C}$ consists two semicircles, the high frequency one contributed by bulk resistance, and low frequency one by grain boundary. The bulk and grain boundary conductivities from equivalent circuit fitting is $1.02 \times 10^{-5} \text{ S/cm}$ and $1.09 \times 10^{-6} \text{ S/cm}$, respectively. It is clear that the grain boundary resistance dominates the total resistance. The reason is that our samples melted between two pieces of gold foil probably were not dense ceramic. Nevertheless, the bulk conductivity is still lower than the previously reported value [20]. In that work [20], the samples were synthesized by heating at $360 \text{ }^{\circ}\text{C}$ under vacuum for long time, resulting in depleted lithium rich anti perovskite $\text{Li}_{3-\delta}\text{OX}_{1-\delta}$. Additionally, Al^{3+} might be doped into the lattice when the sample was melted between Al foil for conductivity test. Lithium vacancies introduced by synthesis and fabrication benefit the conductivity. However, in our work, the samples were synthesized with Li over mount (Li metal, or LiBr in raw materials), so that the as synthesized LiRAP has little Li vacancies. As indicated by theoretical calculation, in LiRAP the nearest-neighbor Li^+ hopping is along the edge of the Li_6O octahedron via Li vacancies [22], and LiRAP with perfect crystal lattice is not good Li^+ conductors [23]. Therefore, based on this established solid state reaction, further efforts of creating Li vacancies in LiRAP are under way.

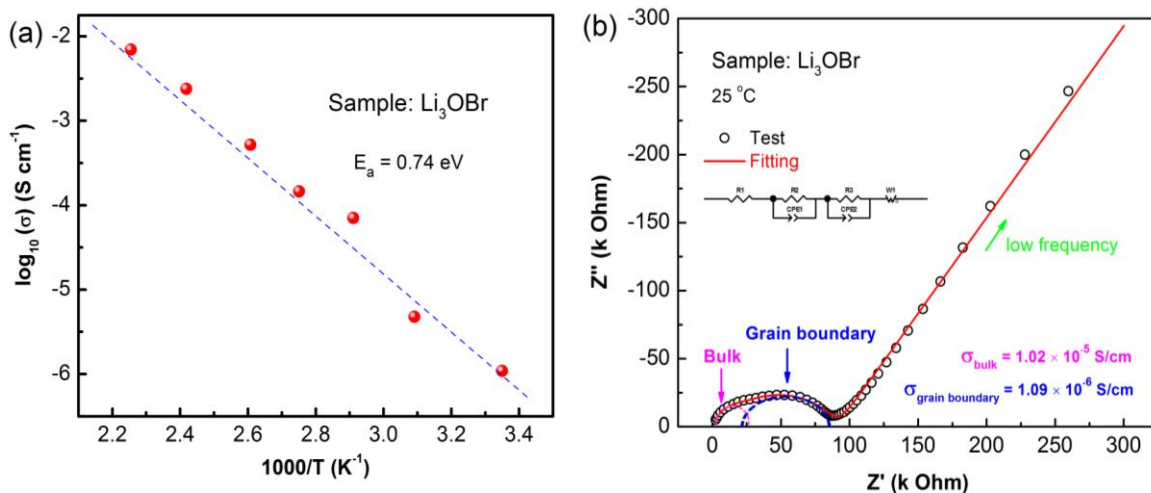


Fig. 6 Arrhenius plot of the total conductivity of Li₃OBr (synthesized from LiBr and Li₂O) between 25 °C and 175 °C with Au foil as electrode; the activation energy E_a is derived by linear fitting of $\ln(\sigma T)$ versus $1/T$. (a) The impedance spectrum of Li₃OBr at 25 °C, and equivalent circuit fitting result. (b)

4. Conclusions

In this work, lithium-rich anti-perovskite Li₃OCl and Li₃OBr were synthesized via solid state reactions. To increase product yield and remove impure phases especially OH phase, different efforts were adopted to improve the synthesis. On one hand, reductive agents such as Li metal or LiH were adopted in starting materials to eliminate the OH phases. The LiOH and LiX form Li₂(OH)X ($X = \text{Cl, Br}$) firstly at relatively low temperature. As temperature increases, Li₂(OH)X melts and reacts with reducer at around 300 °C, forming Li₃OX, releasing H₂ and energy. LiH is more effective than Li metal in the concern of uniform reaction and strong negatively charged H⁻. On the other hand, using Li₂O and LiX as starting materials can form Li₃OX directly and avoiding OH phase at the very beginning. Ball milling can make raw materials to be fine powder, active and contacting thoroughly, therefore, the reaction could be completed by one run of heating with high yield. The advantage of solid state reaction is simple, less requirements on equipments, and easy to scale up. Further studies on creating Li vacancies in LiRAP lattices are under way.

Acknowledgments

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